Task 2: Exploratory Data Analysis (EDA) & Business Intelligence

ApexPlanet Software Pvt. Ltd. Internship | Analysis of Task 1 cleaned sales dataset

1. Objective

Explore the cleaned sales dataset, summarize key business metrics, identify patterns across products, categories, cities and months, and provide SQL questions that can support business decisions.

2. Dataset overview

Item	Result
Rows / columns	1,000 rows / 19 columns
Order date range	2025-01-01 to 2026-01-01
Unique Order_ID values	1,000
Unique customers	947
Missing cells	0
Exact duplicate rows	0
Sales calculation checks failing	0
Rows flagged as sales outliers	19

3. Main KPIs

KPI	Result
Total revenue	■139,399,439.65
Total orders	1,000
Total quantity sold	5,435
Unique customers	947
Average revenue per row/order	■139,399.44
Median sales value	■108,594.02

4. Key findings

• Laptop is the highest-revenue product at ■25,443,008.51; Mobile and Book follow closely.

• Electronics is the highest-revenue category at ■50,778,581.70, accounting for 36.4% of total revenue.

• Patna is the highest-revenue city at ■19,285,966.89.

• The highest revenue month in 2025 is March (■13,059,899.94).

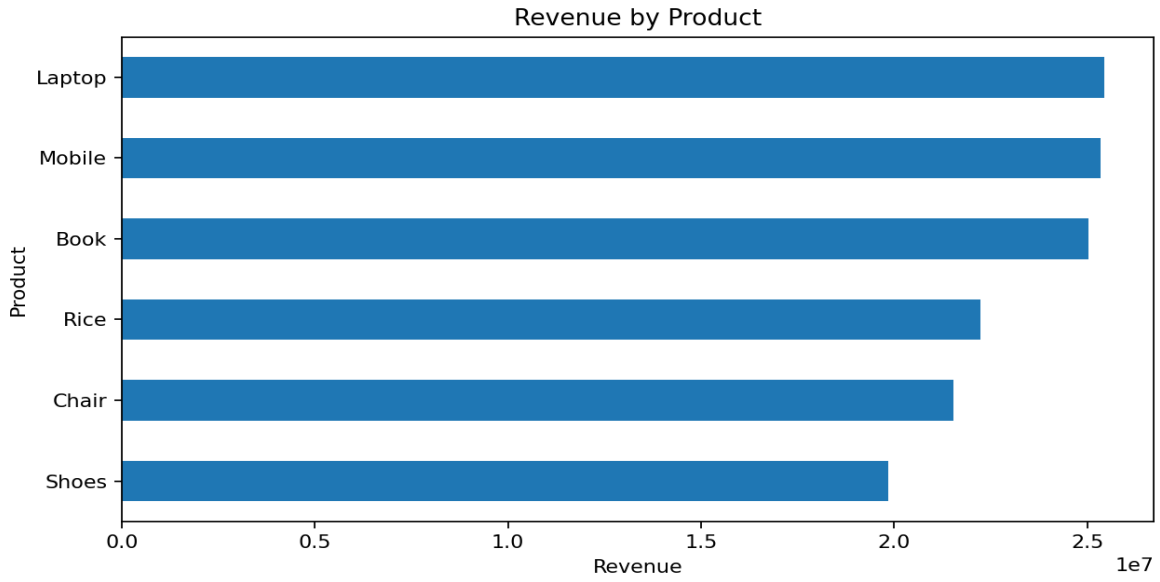
• The 36–50 age group has the highest total revenue (■45,563,823.42).

• All rows have Sales_Calculation_Check = True, with no missing cells and no exact duplicate rows. There are 19 records flagged as potential sales outliers; investigate them rather than deleting automatically.

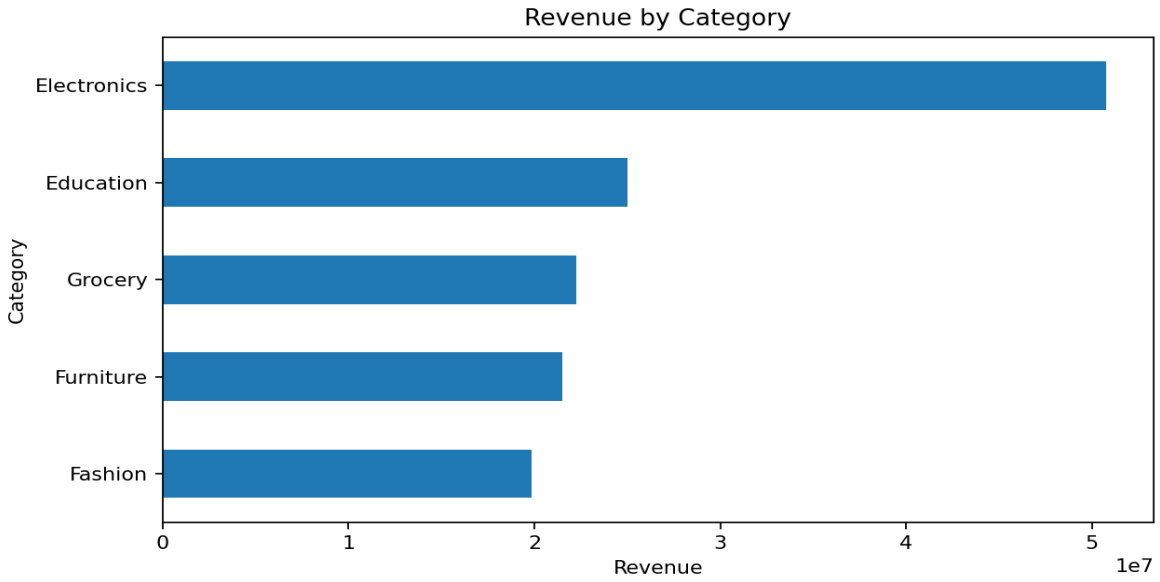
• Order_ID is unique for every row, while Original_Order_ID has repeated values. Keep this distinction in mind when reporting orders.

5. Visual analysis

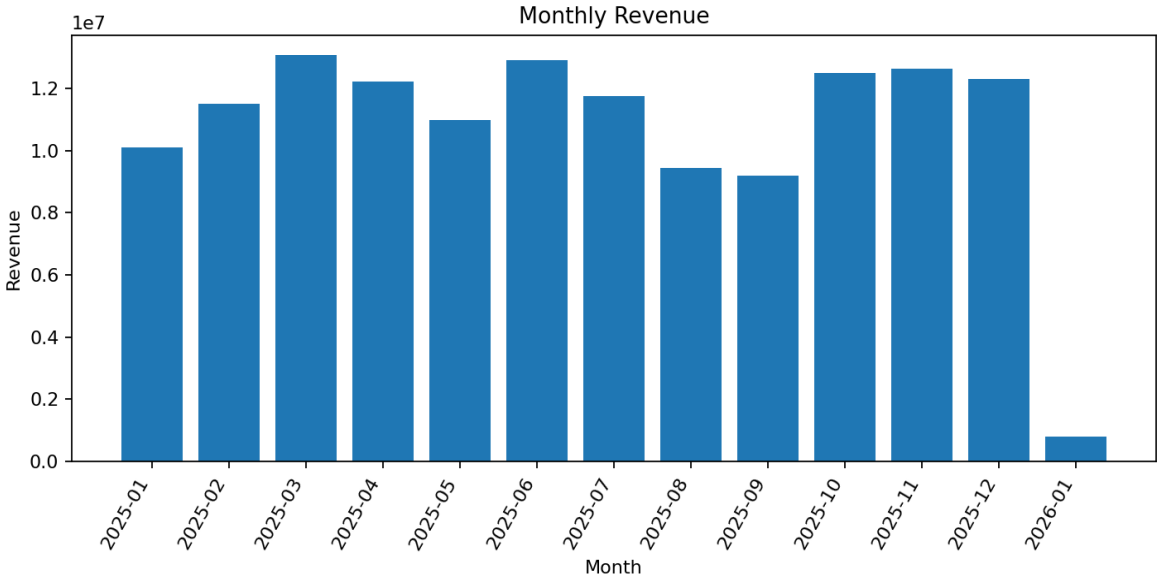
Revenue by product



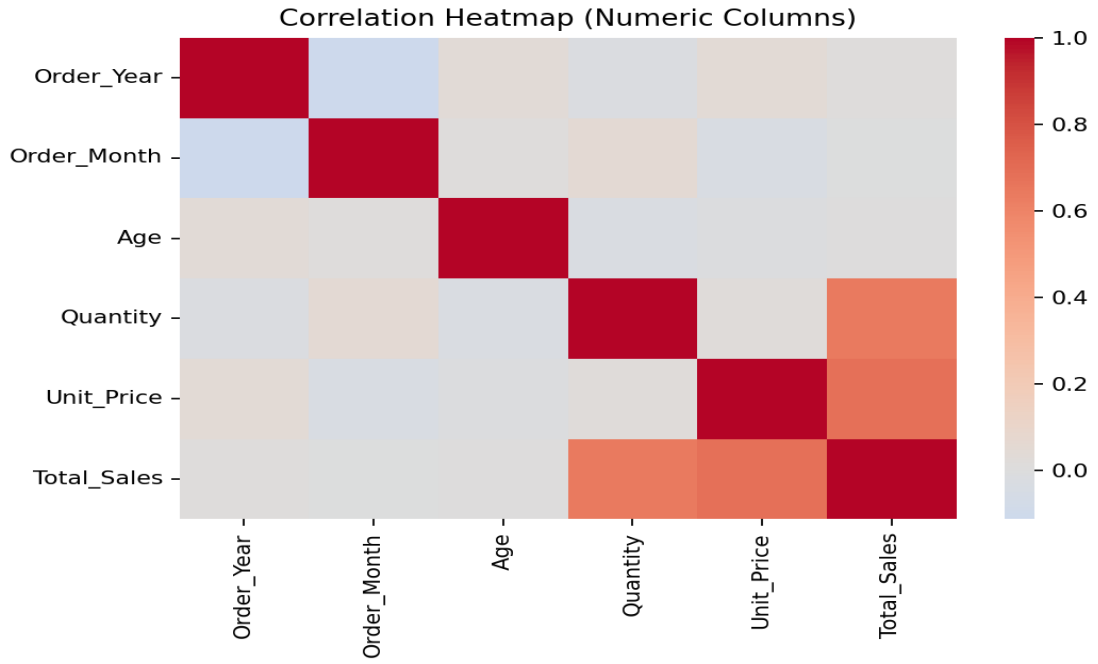
Revenue by category



Monthly revenue



Correlation heatmap



6. SQL business questions

The file task2_queries.sql contains seven queries. The query results are included as separate sheets in task2_dashboard_mockup.xlsx. Queries cover top products, total revenue/orders, category and city performance, monthly revenue, product quantities and age-group revenue.

7. Recommendations

• Review stock and promotions for Laptop and Mobile because they contribute the highest product revenue.

• Study Electronics category performance and maintain availability of high-demand products.

• Compare sales by city and investigate why lower-performing cities differ from Patna and Kolkata.

• Track monthly revenue consistently using both year and month; do not combine January 2025 and January 2026.

• Review the 19 IQR-flagged sales records to check for valid high-value orders or data-entry issues.

8. Conclusion

The dataset contains 1,000 records and 19 columns. The data-quality checks are strong, while the analysis shows differences in product, category, city and monthly performance. The SQL queries and dashboard workbook provide reusable ways to answer business questions. Findings are descriptive and should be checked with business context before making decisions.

Note: This report was generated from the provided cleaned CSV. Currency is displayed with ■ as a presentation format; confirm the source dataset's currency before external business use.